Predicting High vs. Low Loneliness from Smartphone and Wearable Data

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***Abstract*— This study builds on a previously completed study that used smartphone data to predict treatment related mental health outcomes [1]. This study attempts to replicate aspects of that approach in the context of loneliness prediction. Features derived from smartphone behavior, Oura measurements, and UCLA Loneliness Survey responses were used to classify participants into high-loneliness and low-loneliness groups [2]. Four machine learning models were evaluated using 5-fold stratified cross-validation, and XGBoost achieved the strongest overall performance. The results suggest that behavioral and wearable data may be useful for predicting loneliness, although further validation is needed.**

Keywords*— *loneliness, smartphone data, wearable devices, machine learning, behavioral data, UCLA survey

I. Project Goal/Introduction

The goal of this project was to determine whether loneliness could be predicted from behavioral data collected through smartphones and wearable devices. More specifically, the project aimed to classify participants into **high-loneliness** and **low-loneliness** groups using participant-level features derived from smartphone activity and Oura wearable data, with loneliness labels constructed from UCLA Loneliness Survey responses. This work also served as a replication-oriented analysis using data collected in a prior study.

II. Dataset Understanding

A. Dataset Description

The dataset included information from **39 participants** and combined multiple sources of health-related data collected over time [2]. The available inputs included smartphone-based behavioral data, wearable-device data from a smartwatch and an Oura ring, and UCLA Loneliness Survey responses [3]. These different sources made the dataset multimodal, allowing the analysis to capture both behavioral and self-reported aspects of participant well-being.

B. Target Variable

The predicted outcome in this project was **loneliness status**, not depression status. Loneliness-related labels were created from participants' UCLA survey responses. After scoring the UCLA items, each participant received a total loneliness score, which was then converted into two categories: **high_loneliness** and **low_loneliness**. A binary version of the label was also created, where 1 represented high loneliness and 0 represented low loneliness.

C. Initial Observations

Several characteristics of the dataset shaped the analysis. First, it combined different forms of data, including smartphone behavior, wearable measurements, and survey responses. Second, it contained mixed feature types and required aggregation before modeling. Third, missing or incomplete values were present and had to be addressed during preprocessing. In addition, some UCLA survey items were positively worded while others were negatively worded, which required reverse-scoring to ensure consistent interpretation. Finally, the loneliness classes may not have been evenly distributed, making balanced evaluation important.

III. Data Preprocessing

A. Data Cleaning

The preprocessing pipeline handled missing and invalid data mainly through filtering, type standardization, and error skipping. During ETL, CSV files were read in fail-fast mode so malformed files would raise an exception, and unreadable files were skipped rather than allowed to corrupt the dataset. In the survey data, blank or missing responses were filtered out after conversion to long format. The ETL pipeline also cast columns into consistent data types, especially for survey and Oura variables, to reduce schema inconsistencies before modeling.

B. Survey Processing

The UCLA survey data was processed in several steps. First, UCLA rows were selected from the long format survey table, and if a participant had multiple survey instances, only the most recent one was kept. The data was then pivoted so that each participant had one row containing all UCLA question responses. Next, text-based answers were mapped to numeric values using a 4-point scale:

Never = 1, Rarely = 2, Sometimes = 3, Often = 4.

Positively worded UCLA items were reverse scored so that all items pointed in the same direction. Finally, the item scores were summed to create a total UCLA loneliness score for each participant.

After the total score was computed, a median split was used to create two target categories: high_loneliness and low_loneliness. A binary target variable was also created for machine learning, with 1 representing high loneliness and 0 representing low loneliness.

C. Feature Preparation

Features were built from participant level (meaning that each row represents one participant) Aware and Oura measurements. Rather than using raw event level data directly, the project aggregated smartphone and wearable records into

summary statistics such as counts, averages, and totals. Examples included counts of calls, messages, notifications, and screen events, as well as average sleep, activity, and readiness measures from Oura data. After merging these feature tables, identifier and label-related columns were excluded from the modeling inputs. Missing numeric feature values were then filled with the median of each column, producing a participant-level dataset suitable for classification.

D. Train/Test Preparation

After the participant level dataset was created, the data was separated into predictors (X) and the binary loneliness target (y). The **feature matrix** excluded participant identifiers and UCLA label columns, while loneliness_label_binary served as the **target variable**. Model evaluation was performed using 5-fold stratified cross-validation with shuffling and a fixed random state, ensuring that each fold maintained a similar class distribution [4].

IV. Machine Learning Methods

A. Models Used

The project evaluated four classification models: **Support Vector Machine (SVM), Random Forest, XGBoost, and a Neural Network** [5][6][7][8]. SVM was included as a strong nonlinear classifier for structured data. Random Forest and XGBoost were selected because they are effective ensemble methods for tabular datasets and can capture complex relationships among features. A neural network was also included to compare a more flexible learning approach with the more traditional machine learning models.

B. Training Procedure

All models were trained and evaluated using 5-fold stratified cross-validation. In this procedure, the dataset was split into five folds, and each model was trained and tested five times so that each fold served once as a validation set. This approach helped preserve the proportion of high- and low-loneliness participants across folds and provided a more reliable estimate of model performance.

Each model was implemented as part of a preprocessing pipeline. All four pipelines used median imputation to handle missing values. The SVM and Neural Network pipelines also applied standardization, while Random Forest and XGBoost used imputation only, since tree-based models do not require feature scaling in the same way.

C. Evaluation Metrics

The models were evaluated using **accuracy, precision, recall, F1 score, and ROC-AUC**. Accuracy measures the overall proportion of correct predictions. Precision measures how many predicted positive cases were actually positive.

Recall measures how many actual positive cases were correctly identified. F1 score balances precision and recall into a single measure, and ROC-AUC evaluates the model's ability to distinguish between classes across thresholds. Multiple metrics were necessary because accuracy alone can be misleading, especially when the class distribution is uneven. Together, these metrics provided a fuller view of model performance.

V. Results

A. Cross-Validation Performance

Cross-validation results showed that XGBoost achieved the strongest overall performance (Fig 1). Across the five folds, it obtained an average accuracy of 0.70, precision of 0.70, recall of 1.00, and F1 score of 0.80. These results suggest that XGBoost was the most effective model for identifying high-loneliness participants in this dataset.

The remaining models performed more modestly. The Neural Network achieved an average accuracy of 0.50, but its precision (0.10), recall (0.20), and F1 score (0.13) were low. Random Forest had an average accuracy of 0.40, precision of 0.10, recall of 0.20, and F1 score of 0.13. SVM performed worst overall, with an accuracy of 0.30, precision of 0.20, recall of 0.40, and F1 score of 0.27. Although Random Forest produced the highest reported ROC-AUC (0.80), its threshold-based classification performance remained weak.

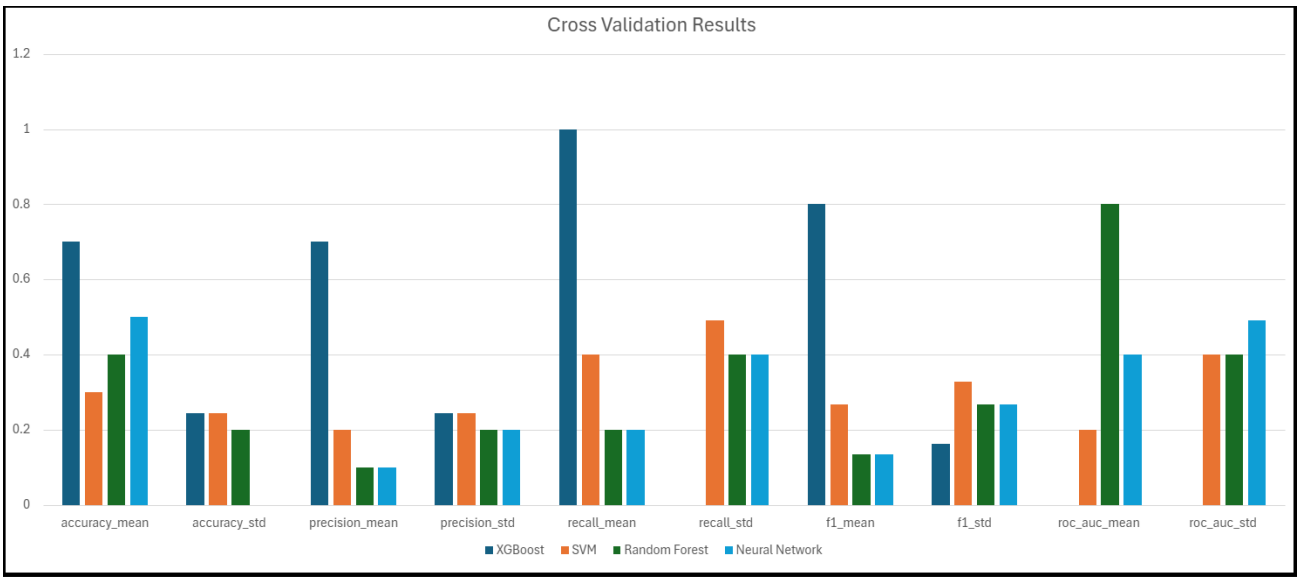

Fig 1: XGBoost performs best overall and achieves the strongest predictions results. While Random Forest did not perform as well, it provided a more balanced and interpretable feature importance profile.

B. Model Comparison

Among the four models, **XGBoost** was the best-performing model overall. It had the highest accuracy, precision, recall, and F1 score, which means it provided the strongest overall balance and was especially effective at detecting high-loneliness cases. Its recall of 1.00 was particularly notable, since it indicates that all positive cases were identified in cross-validation.

Random Forest and the **Neural Network** showed weaker overall balance, especially in terms of precision, recall, and F1 score. However, Random Forest stood out by achieving the strongest ROC-AUC, suggesting that it may still have had some ability to separate the two classes even though its classification threshold did not produce strong final predictions. **SVM** showed the weakest overall performance and appeared least suitable for this task.

C. Key Findings

The main numerical finding of the study was that XGBoost substantially outperformed the other models, achieving an average accuracy of 0.70, precision of 0.70, recall of 1.00, and F1 score of 0.80. Random Forest achieved the highest reported ROC-AUC at 0.80, but its precision, recall, and F1 score were low. Overall, these results suggest that smartphone and wearable derived behavioral features were useful predictors of loneliness, although the fact that only one model performed strongly suggests that the predictive signal may have been limited or unstable.

VI. Discussion

A. Interpretation of Results

In practical terms, the results suggest that behavioral and wearable data contained meaningful information related to loneliness, but the usefulness of that information depended strongly on the model used. XGBoost performed well enough to indicate that the data captured learnable patterns associated with high loneliness. At the same time, the weaker performance of the other models suggests that those patterns were not consistently easy to detect. For that reason, the results are best interpreted as promising but limited. The project demonstrated that loneliness could be predicted to some extent from smartphone and wearable behavior, but the evidence was not uniformly strong across all models.

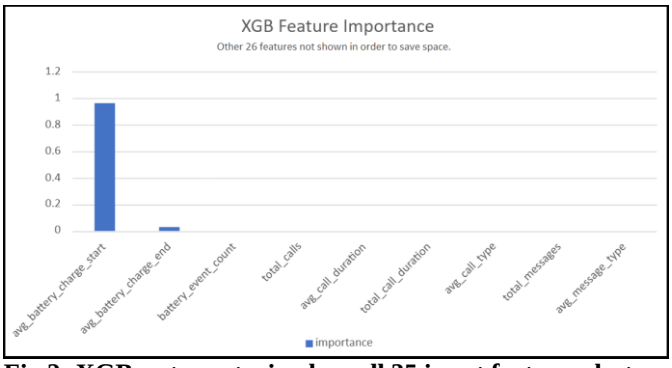

Fig 2: XGBoost was trained on all 35 input features, but only two features received nonzero importance. Phone charging timing was the strongest, while the rest of the features added little predictive value. Perhaps phone charging is acting as a proxy for sleep schedule or phone usage routine.

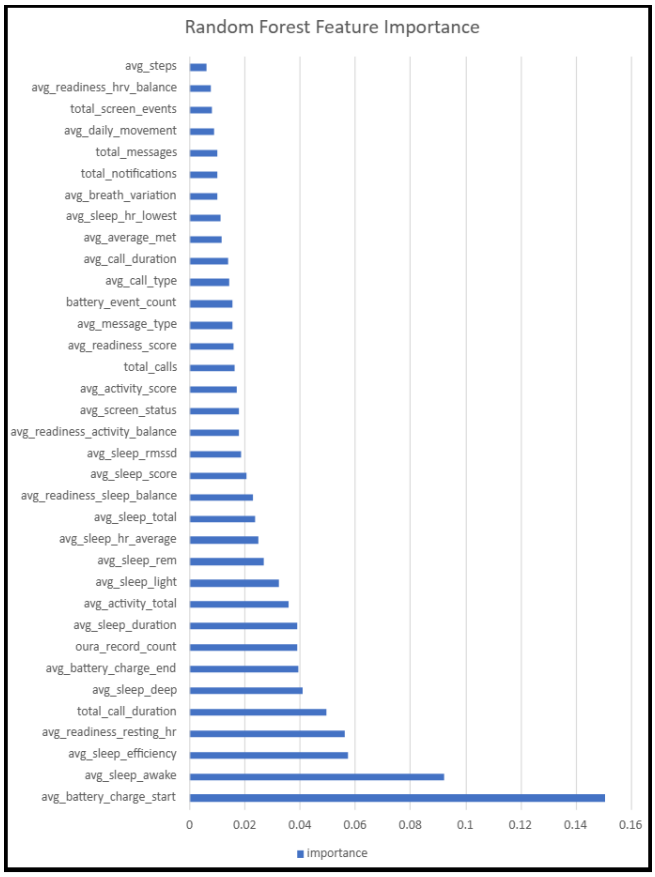

Fig 3: The features "average battery charge start" and "average sleep awake" appeared to be the most important predictors, suggesting that daily phone charging and sleep related routines may have better distinguishing high and low loneliness participants than general activity level.

B. Strengths of the Approach

This approach had several strengths. First, it integrated multiple data sources, including smartphone behavior, wearable measurements, and survey-based loneliness labels. Second, the preprocessing pipeline was careful and systematic, including response mapping, reverse-scoring, participant-level feature aggregation, and median imputation. Third, model evaluation used stratified 5-fold cross-validation, which helped preserve class balance across folds. Finally, the models were assessed with multiple evaluation metrics, which provided a more complete and reliable view of performance than accuracy alone.

C. Limitations

The project also had several limitations. One likely limitation was the small participant level dataset, which may have contributed to instability across folds and across models. Missing or incomplete data were also an issue. Although preprocessing and imputation improved data quality, skipped malformed files and incomplete observations may still have reduced the amount of usable information. Another limitation was the use of a median split to convert a continuous UCLA loneliness score into a binary target. While convenient for classification, this approach reduces nuance and may oversimplify real variation in loneliness. Finally, because only XGBoost performed strongly, the results should be interpreted cautiously and may not generalize without further validation.

D. Future Work

Future work should focus on expanding the dataset, improving feature engineering, and testing the models on independent data. More participants and more complete records would likely improve model stability. In addition, richer time based behavioral features, such as day-to-day variability in sleep, communication, and phone usage, may be more informative than simple averages and counts. Future studies could also preserve more information by treating loneliness as a continuous or multi-level outcome rather than reducing it to a binary label. These improvements would help determine whether the results generalize beyond the current dataset and whether the predictive signal remains strong in a broader setting.

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